Skills Summary

Slope and Rate of Change

Use this reference while completing your worksheet or when studying.

One idea, four places to find it. A rate of change says how much y moves for each 1 that x moves. Written as slope: $m = \frac{y_2 - y_1}{x_2 - x_1}$.

Graph: two lattice points. **Table:** two rows. **Equation** $y = mx + b$**:** the number m in front of x. **Story:** the “per” number.

A *positive* rate rises, a *negative* rate falls, and the size of the number is how steep it is.

1. Slope from two points

Label the points (x_1, y_1) and (x_2, y_2) *before* subtracting, and keep that order on the top and the bottom. Swapping one and not the other flips the sign of your answer. On a graph, use points the line passes exactly through.

Example: Find the slope through $(1, 4)$ and $(5, 16)$.
Step 1. Two points and nothing else, so this is the slope formula; the only real decision is which point is which, and then keeping it.
Step 2. $m = \frac{16-4}{5-1} = \frac{12}{4}$
$\Rightarrow$ **3** (y rises 3 for every 1 that x rises)

Try it: Find the slope through $(2, 9)$ and $(5, 3)$.

check: **−2**

2. Rate of change from a table

Pick any two rows and divide the change in the output by the change in the input. If every pair of rows gives the *same* answer, the table is linear and that number is its rate; if two intervals disagree, there is no single rate — which is exactly how you prove a table is not linear.

Example: Find the rate of change of this table.

x	1	3
y	7	15

Step 1. A table gives points, so the rate is a slope — read one column as $(1, 7)$ and the other as $(3, 15)$.
Step 2. $m = \frac{15-7}{3-1} = \frac{8}{2}$
$\Rightarrow$ **4** per unit of x

Try it: $x = 0$ gives $y = 20$; $x = 2$ gives $y = 14$. What is the rate?

check: **−3**

3. Rate of change from an equation

Get the rule into the form $y = mx + b$. Then m is the rate and b is only the starting value — b never affects the rate, because it is added once and never changes. Not sure? Evaluate at $x = 0$ and $x = 1$; the jump between those two outputs is m.

Example: What is the rate of change of $y = 6x - 4$?
Step 1. The rule is already in $y = mx + b$ form, so the rate is read off, not computed — but check it once with two points to be sure.
Step 2. At $x = 0$, $y = -4$; at $x = 1$, $y = 2$. The jump is $2 - (-4) = 6$.
$\Rightarrow$ **6** (the -4 is the starting value, not part of the rate)

Try it: What is the rate of change of $y = -7x + 5$?

check: -7

4. Comparing two rates

You cannot compare a table with an equation directly. Turn *each* one into a rate in the same units first, then compare the two numbers. With negatives, compare positions on a number line: -5 is less than -2, even though -5 is the steeper fall.

Example: Plan A costs $y = 9x$ dollars for x hours. Plan B costs 33 dollars for 3 hours. Which rate is larger?
Step 1. The two plans are given in different forms, so neither can be compared until both are a cost per hour.
Step 2. Plan A: the coefficient gives 9 per hour. Plan B: $33 \div 3 = 11$ per hour. Now compare 9 with 11.
$\Rightarrow$ Plan A's rate $<$ Plan B's rate

Try it: One rate is 15 per hour, the other is 12 per hour. Write $<$ or $>$: first ____ second.

check: $>$

(!) Watch out: bigger numbers in a table do not mean a bigger rate — they may just cover more hours. And when rates include negatives, "fastest" and "greatest" are different questions: -5 is the fastest fall and the smallest number.